



University of Asia Pacific
Department of Computer Science and Engineering
CSE 450: VLSI Lab

LAB REPORT

Experiment Number: 01

Experiment Title: Design and Simulation of a Two-Input CMOS NOR Gate Using Cadence Virtuoso

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1. Experiment Name

Design and Simulation of a Two-Input CMOS NOR Gate Using Cadence Virtuoso.

2. Objective

The main objectives of this experiment are as follows:

- To understand the working principle and Boolean logic of a two-input CMOS NOR gate.
- To design the NOR gate at the transistor level using complementary PMOS and NMOS networks in the Cadence Virtuoso Schematic Editor.
- To create a re-usable schematic symbol for the designed NOR gate using the Symbol Editor.
- To build a test-bench circuit by instantiating the NOR gate symbol and driving its inputs with independent pulse voltage sources.
- To set up and run a transient (time-domain) simulation using the Spectre simulator through ADE.
- To capture and analyze the input–output waveforms in the ViVA Waveform Viewer and verify them against the ideal NOR truth table.

3. Apparatus / Hardware & Software Requirements

The following software tools, libraries, and hardware resources were used to carry out the design and simulation:

Item	Description / Specification
EDA Tool	Cadence Virtuoso Studio — Schematic Editor & Symbol Editor
Technology Library	GPDK045 — 45 nm generic process design kit (svt devices)
Simulator	Spectre Circuit Simulator, invoked through ADE (Analog Design Environment)
Waveform Viewer	ViVA Graph — for transient response visualization
Computing Platform	Linux workstation with a valid Cadence license (vlsicad server)
Version Control	Git & GitHub — for tracking and submitting design files

4. Circuit Diagram / Schematic

4.1 Working Principle

A NOR gate produces a logic HIGH output only when all of its inputs are LOW; if any input is HIGH, the output is LOW. For two inputs A and B, the Boolean expression is:

$$Y = (A + B)' = A' \cdot B'$$

In static CMOS logic, this function is implemented using two complementary networks connected between the supply rails: a Pull-Up Network (PUN) made of PMOS transistors that connects the output to VDD, and a Pull-Down Network (PDN) made of NMOS transistors that connects the output to GND. Because the NOR expression is an OR followed by inversion, the PDN consists of NMOS transistors in parallel (so the output is pulled LOW if either A or B is HIGH), while the PUN consists of PMOS transistors in series (so the output is pulled HIGH only when both A and B are LOW). This complementary structure guarantees that exactly one of the two networks conducts at any time, giving rail-to-rail logic levels with no static power dissipation.

4.2 Transistor-Level Schematic

Figure 1 shows the transistor-level implementation of the two-input CMOS NOR gate designed in the Cadence Virtuoso Schematic Editor. It consists of two PMOS transistors (PM0, PM1) connected in series between VDD and the output node Vout, and two NMOS transistors (NM0, NM1) connected in parallel between Vout and ground. Inputs Va and Vb drive the gate terminals of the corresponding PMOS–NMOS transistor pairs.

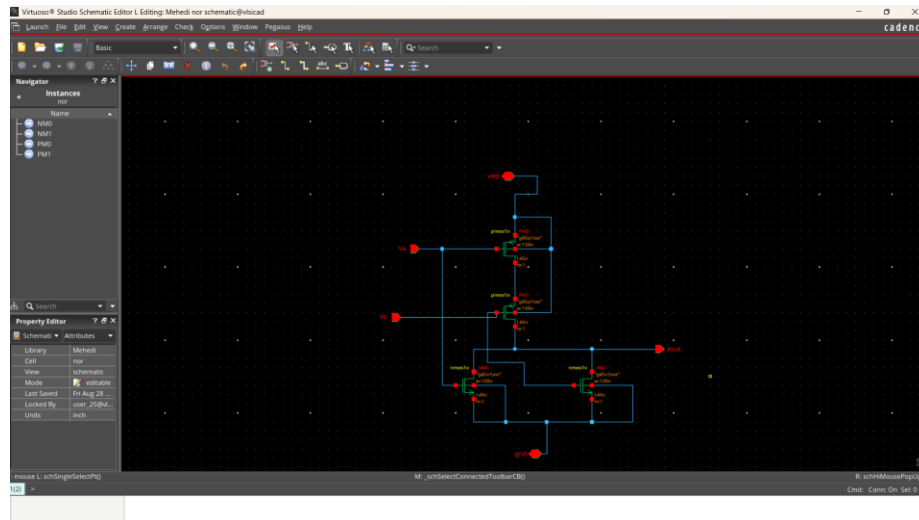


Figure 1: Transistor-level schematic of the CMOS NOR gate (Cadence Virtuoso)

The sizing of the four transistors used in the design is summarized in Table 1. Standard-threshold-voltage (svt) devices from the GPDK045 library were used for both the PMOS and NMOS transistors.

Transistor	Type	Width (W)	Length (L)
PM0, PM1	PMOS (g45p1svt)	120 nm	45 nm
NM0, NM1	NMOS (g45n1svt)	120 nm	45 nm

Table 1: Transistor sizing used in the CMOS NOR gate design

4.3 Schematic Symbol

After verifying the transistor-level circuit, a schematic symbol was created for the NOR gate so that it could be reused as a single instance in higher-level schematics. Figure 2 shows the resulting symbol, with input pins Va and Vb, output pin Vout, and the supply pins vdd and gnd.

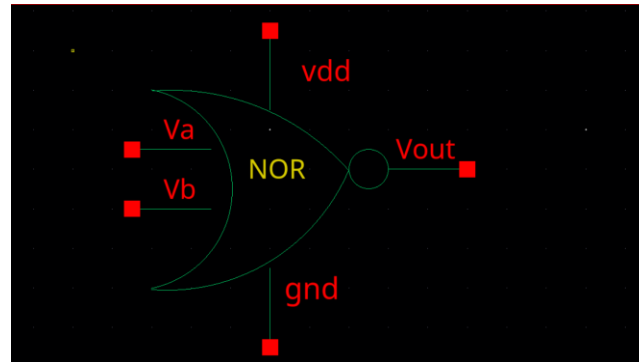


Figure 2: Schematic symbol of the designed NOR gate

4.4 Test-Bench Schematic

Figure 3 shows the test-bench schematic built for simulation. The NOR gate symbol is instantiated once, and its two inputs are driven independently so that all four input combinations occur over the simulation window:

- Va is driven by pulse source V0 (0 V to 1 V, rise/fall time $t_r = 50$ ps).
- Vb is driven by pulse source V1 (0 V to 1 V, rise/fall time $t_r = 50$ ps, different pulse period from V0 so all A–B combinations appear).
- The vdd pin is connected to a 1 V DC source (V2), and the gnd pin is tied to the global ground.

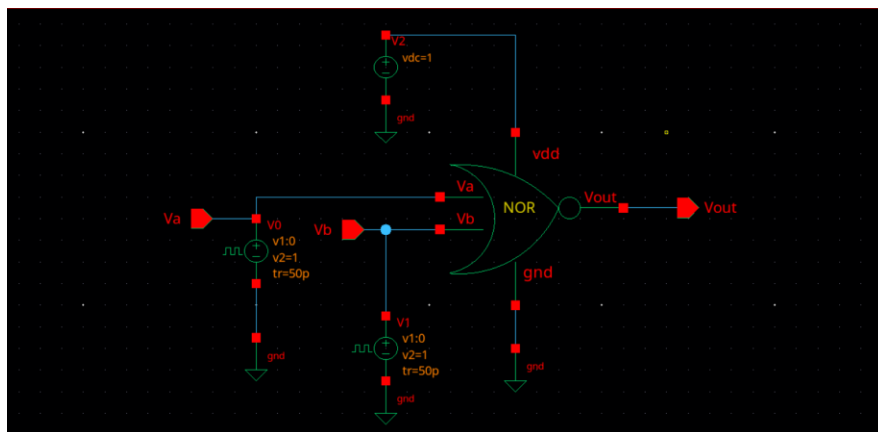


Figure 3: Test-bench schematic with pulse sources for transient simulation

6. Output Analysis / Observations

6.1 Simulation Setup

A transient (tran) analysis was configured in ADE with a simulation time span of 0 to 100 ns, using the Spectre simulator. The node voltages Va, Vb, and Vout were selected for plotting, and the results were viewed in the ViVA Waveform Viewer.

6.2 Waveform Observation

Figure 4 shows the captured transient response. The red trace (/Y) is the output Vout, the green trace (/A) is input Va, and the magenta trace (/B) is input Vb, all plotted on a 0–100 ns time axis.

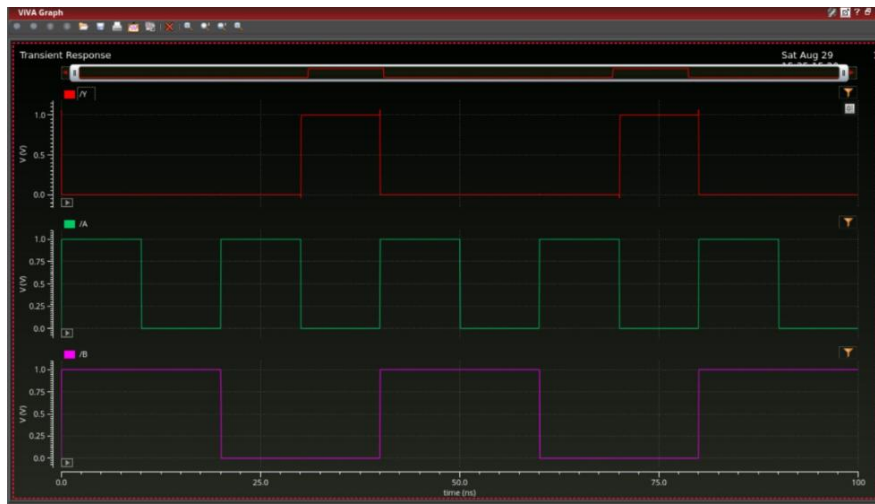


Figure 4: Transient response of the CMOS NOR gate

Careful inspection of the waveform shows that:

- When both A and B are LOW (0 V), the output Y rises to HIGH (≈ 1 V) — visible as the wide red pulses that align exactly with the low regions of both the green and magenta traces.
- As soon as either A or B switches HIGH, the output Y immediately falls back to LOW (0 V) and stays LOW until both inputs return to 0 V again.
- The output transitions occur sharply, with fast rise and fall times consistent with the 50 ps input transition time, confirming correct switching behaviour of the PUN/PDN networks.

This behaviour exactly matches the truth table of a two-input NOR gate, confirming the correct logical functionality of the designed circuit:

A (Va)	B (Vb)	Y = A NOR B (Vout)
0	0	1
0	1	0
1	0	0
1	1	0

Table 2: Truth table of the two-input NOR gate, verified against the simulated waveform

7. Result

A two-input CMOS NOR gate was successfully designed at the transistor level in Cadence Virtuoso using a complementary Pull-Up Network of series-connected PMOS transistors and a Pull-Down Network of parallel-connected NMOS transistors. A schematic symbol was created and instantiated in a dedicated test-bench, and a transient simulation was carried out using the Spectre simulator. The simulated output waveform matched the ideal NOR truth table for all four input combinations (00, 01, 10, 11), with the output going HIGH only when both inputs were LOW. The design shows sharp, rail-to-rail logic transitions and correct switching behaviour, confirming that the objectives of the experiment were fully achieved.

8. GitHub Repository Commits

All design files, schematics, and the commit history for this lab have been pushed to the following personal GitHub repository:

https://github.com/mehedihasan22201109/VLSI_LAB

Click on the link above (Ctrl + Click in Word) to view the repository and the commit history for this experiment.

9. Conclusion

This experiment provided hands-on experience with the complete static CMOS gate-design flow — from schematic capture to functional verification. Working through the design, the following key points were learned:

- How a complementary Pull-Up Network (series PMOS) and Pull-Down Network (parallel NMOS) together realize the NOR Boolean function with full rail-to-rail swing.
- How to build and connect a transistor-level schematic, and how to create a corresponding schematic symbol for hierarchical reuse in Cadence Virtuoso.
- How to construct a test-bench with independent pulse sources to exercise all input combinations of a logic gate.
- How to configure and run a Spectre transient simulation through ADE, and how to interpret the resulting waveforms in ViVA to verify digital functionality against a truth table.

Overall, the designed NOR gate behaved exactly as expected, and this exercise has strengthened the practical understanding of static CMOS logic design, forming a solid foundation for designing and simulating more complex digital circuits in future experiments.